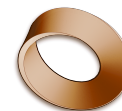


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.6 – Sound Waves

Unit 1: Oscillations & Waves

Learning intentions

- a) I can describe how vibrating objects produce **sound** waves through particle interactions.
- b) I can explain why sound requires a **medium** and cannot travel through a vacuum.
- c) I can distinguish between **pitch** (frequency) and **loudness** (amplitude) in sound waves.
- d) I can calculate the speed of sound using the relationship $v = f\lambda$.

Theory

Sound is a mechanical wave produced when an object vibrates, such as a guitar string or vocal cords. As the object moves forward, it pushes nearby air particles together, creating regions of high pressure called **compressions**. When the object moves backward, it leaves behind regions where particles are spread apart, known as **rarefactions**. Because the particle oscillations occur parallel to the direction of energy transfer, sound is classified as a **longitudinal** wave.

Unlike light, which can travel through empty space, sound needs a material **medium** to propagate. This is because sound relies on particles bumping into one another to pass energy along. If you place an electric bell inside a sealed glass jar and pump the air out to create a vacuum, the sound becomes fainter and eventually inaudible, even though you can still see the bell vibrating. This **bell-jar demonstration** proves that without particles to transmit the vibrations, sound cannot travel.

The characteristics of a sound wave are defined by its frequency and amplitude. **Pitch** refers to how high or low a sound seems to our ears and depends directly on the **frequency** (f), which is the number of complete wave cycles passing a point per second, measured in hertz (Hz). A higher frequency results in a higher pitch. **Loudness**, on the other hand, depends on the **amplitude** of the wave, which represents the maximum displacement of particles from their rest position. Larger amplitude means more energy is carried by the wave, resulting in a louder sound.

The human ear can typically detect frequencies between 20 Hz and 20 000 Hz. Sounds with frequencies above 20 000 Hz are called **ultrasound**. The speed of sound varies depending on the medium; in air at room temperature, it is approximately 330 m/s. When sound waves hit a hard surface, they bounce back as an **echo**, which is simply a reflected sound wave.

To calculate the speed of a sound wave, we use the wave equation: speed = frequency $\times$ wavelength
 $v = f\lambda$

Worked Example: A tuning fork vibrates at a frequency of 250 Hz. The wavelength of the sound produced is measured to be 1.32 m. Calculate the speed of this sound wave. 1. Identify the known values: $f = 250$ Hz and $\lambda = 1.32$ m. 2. State the formula: $v = f \times \lambda$. 3. Substitute the values: $v = 250 \times 1.32$. 4. Calculate the result: $v = 330$ m/s.

Complete the sentence: Sound is a longitudinal wave that requires a _____ to travel through, meaning it cannot propagate in a vacuum.

Word bank: medium · frequency · amplitude · echo · ultrasound

Activity

Investigating Echoes and Speed of Sound

You will need: A stopwatch (or phone timer), a large flat wall or cliff face, and a measuring tape (if available). If you cannot measure the distance to a wall, assume the distance is 50 m.

- Stand facing a large, hard surface (like a school hall wall or a building) at a known distance d .
- Clap your hands sharply once and start the stopwatch immediately.
- Stop the watch the moment you hear the echo return.
- Record the time t in the table below.
- Repeat the measurement two more times to find an average time.

Trial	Time (s)
1	_____
2	_____
3	_____
Average	_____

Note: The sound travels to the wall and back, so the total distance is $2d$. Use the formula $v = \frac{2d}{t}$ to calculate your experimental speed of sound.

Questions

1. Explain why you cannot hear a bell ringing inside a vacuum chamber, even though you can see it vibrating.

2. If the frequency of a sound wave increases, what happens to its pitch?

3. A sound wave has a frequency of 440 Hz and a wavelength of 0.75 m. Calculate the speed of this sound wave.

4. Why is ultrasound inaudible to humans?

5. If you double the amplitude of a sound wave, how does the loudness change?

Extension

Research the principle of **sonar** (Sound Navigation and Ranging). Explain how submarines or fish finders use the time delay between sending an ultrasound pulse and receiving its echo to calculate the depth of the sea floor or the distance to a school of fish.
